Compressible Tensor Language Models: A structured MPO Representation for Large Language Models

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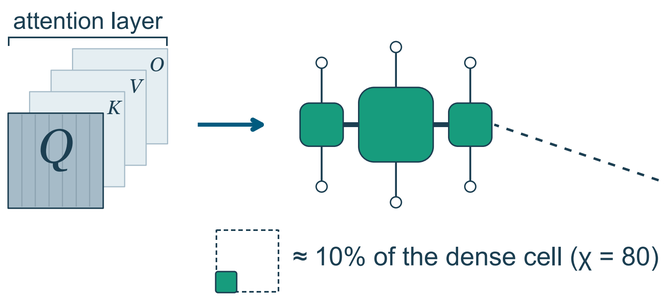
SCAN ME

Abstract

Conventional low-rank methods offer us a single rank constraint on each weight matrix. They are efficient when low rank is sufficient, but their parameter-efficiency advantage diminishes as the required rank for specific tasks increases. Matrix product operators (MPOs) provide a structured factorization in which selected weight matrices are represented as chains of local tensor cores. Compression is controlled by the bond dimension "χ" , which determines the expressive capacity of the MPO representation and enables a block-layer truncation policy based on model sensitivity.

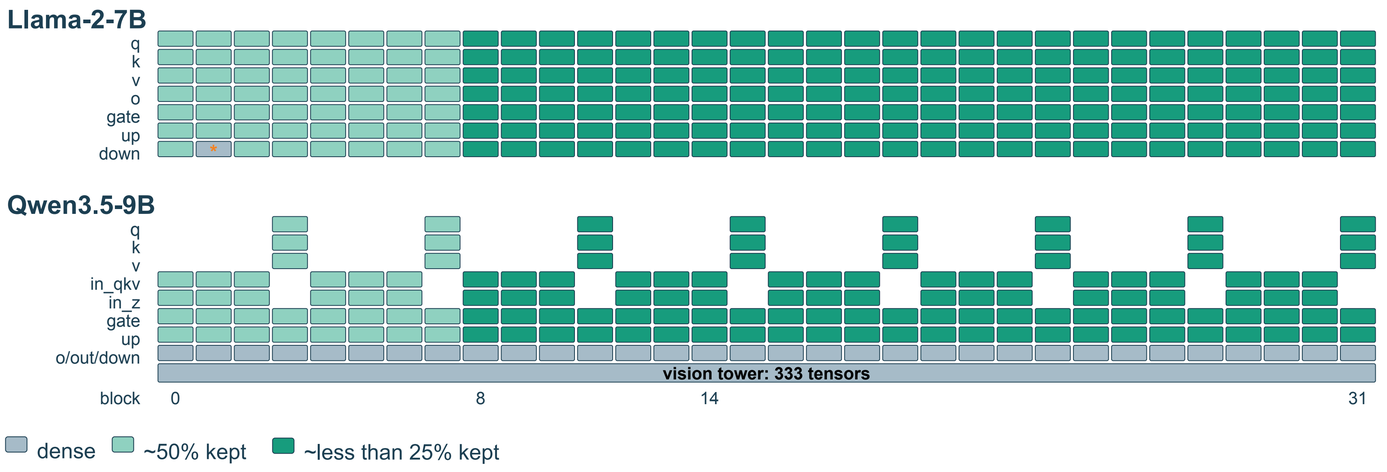
We study this structure across several LLMs and perform an end-to-end MPO compression, retraining and benchmark evaluation on Llama2-7B-chat. The resulting model has a reduced parameter count and model-weight memory footprint. Applying the truncation policy directly severely degrades its next token prediction and downstream benchmark performance. Nevertheless, retraining only the compressed components of the model, which represent 4.4% of the original model's parameters, we observe that it substantially recovers perplexity (PPL), as well as its LLM benchmarks. This recovery required 6.5 hours on a single A100-40GB GPU, compared with 184,320 A100-80GB GPU-hours reported for Llama-2-7B pretraining. The resulting model achieves the same level of performance as the original model with fewer parameters and a smaller model weight memory footprint.

Dense or structured?



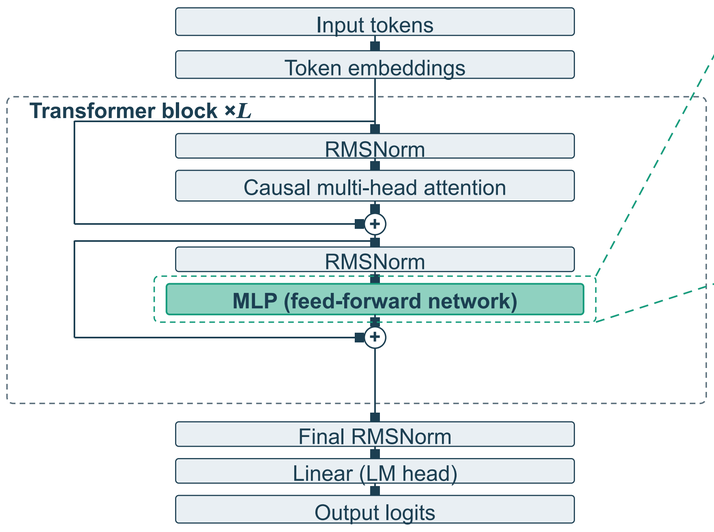
- Scaling explains why bigger models work, not why the dense weights are efficiently structured.
- A matrix product operator[4–7] replaces a weight matrix by a chain of small cores.
- The bond dimension χ sets its capacity.

Model Sensitivity Profiling

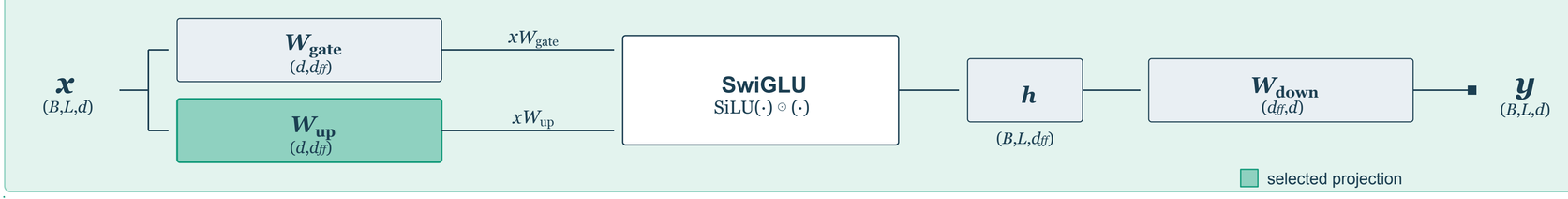


One down-projection super-weight cell stays dense*[9]. Early blocks are the most fragile parts, so recoveryneeds a prior analysis

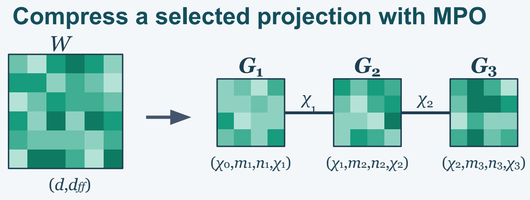
MPO Compression Pipeline



Example: MLP in a Llama-style block



Compress a selected projection with MPO



- $W_{up} \approx \mathrm{MPO}(G_1, G_2, G_3)$
- $W(i,j) \approx G_1[i_1, j_1] \cdot G_2[i_2, j_2] \cdots G_d[i_d, j_d]$
- Parameters: $\Sigma_k \chi_{k-1} m_k n_k \chi_k \ll d \cdot d_{ff}$

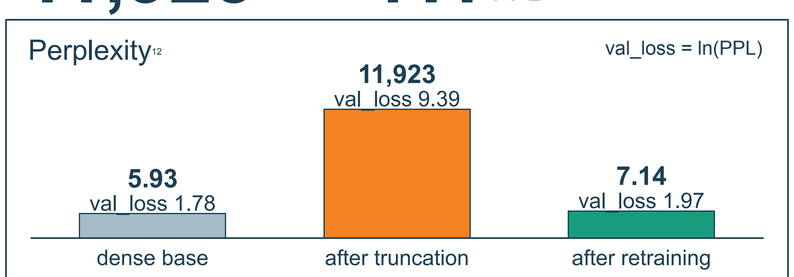
Retrain tensor cores

- **Trainable:** the tensor cores of each compressed projection.
- **Frozen:** embeddings, norms, uncompressed projections, LM head.
- **Objective:** next-token loss; gradients reach only the cores.

Some figures were created with the assistance of generative AI tools. All reported content has been reviewed and validated

Performance Results - Benchmarks

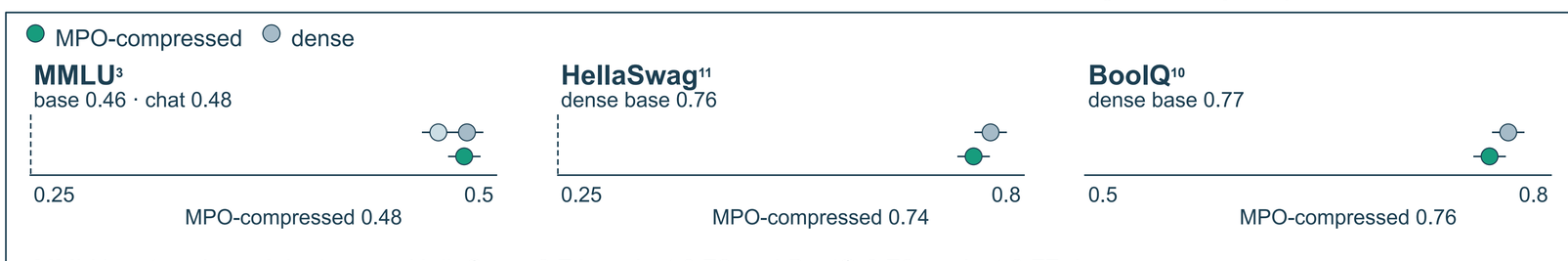
11,923 → 7.1 PPL



Perplexity recovers to 1.2× the dense value.

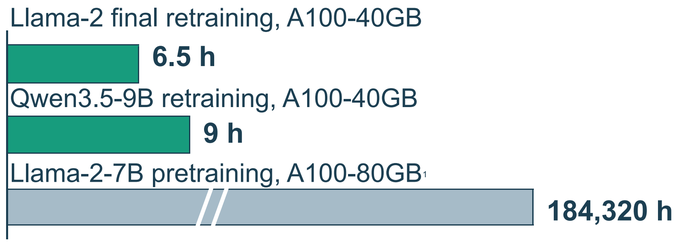
0.48 MMLU score

−67% weight memory footprint



MMLU restored to original score, HellaSwag 0.74 against 0.76 and BoolQ 0.76 against 0.77 dense score

Computational cost



GPU-hours **≈ 10⁴× fewer**

- Llama-2: 157 M tokens. Stored bf16 weights: 4.4 GB against 13.5 GB (not runtime memory).

Latest compressed model: Qwen3.5-9B

698
tensors bit-identical:365 text + 333 vision

7,543 → 16.9
Perplexity

9 GPU-h
Peak 28.6 GiB on one A100-40GB

0.77
MMLU score

MEASURED	dense 9B	before retraining	retrained
Perplexity[12]	8.32	7,543	16.9
HellaSwag[11]	0.71	0.31	0.70
BoolQ[10]	0.89	0.38	0.86
MMLU probe[3], 900 items, 5-shot	0.80	0.25	0.77
Parameters, text model	8.95 B	5.81 B	5.81 B (−35.1%)
Memory, bf16 weights	17.9 GB	11.6 GB	11.6 GB

References

[1] Touvron et al., Llama 2: Open Foundation and Fine-Tuned Chat Models, arXiv:2307.09288 (2023).
[2] Oseledets, Tensor-Train Decomposition, SIAM J. Sci. Comput. 33, 2295 (2011).
[3] Hendrycks et al., Measuring Massive Multitask Language Understanding, ICLR 2021.
[4] Schollwöck, The density-matrix renormalization group in the age of matrix product states, Ann. Phys. 326, 96 (2011).
[5] Tomut et al., CompactifAI: Extreme Compression of Large Language Models using Quantum-Inspired Tensor Networks, arXiv:2401.14109 (2024).
[6] Novikov, Podoprikhin, Osokin, Vetrov, Tensorizing Neural Networks, NeurIPS 28, 442 (2015).
[7] Gao et al., Compressing deep neural networks by matrix product operators, Phys. Rev. Research 2, 023300 (2020).
[8] Liu, Gao, Zhao, Xie, Lu, Wen, Enabling Lightweight Fine-tuning for Pre-trained Language Model Compression based on Matrix Product Operators, ACL 2021, 5388-5398.
[9] Yu et al., The Super Weight in Large Language Models, arXiv:2411.07191 (2024).
[10] Clark et al., BoolQ: Exploring the Surprising Difficulty of Natural Yes/No Questions, NAACL 2019, 2924-2936.
[11] Zellers et al., HellaSwag: Can a Machine Really Finish Your Sentence?, ACL 2019, 4791-4800.
[12] Merity, Xiong, Bradbury, Socher, Pointer Sentinel Mixture Models, ICLR 2017.

Acknowledgements

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